Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *ARABICMMLU* | ***Context:*** *Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark is uniformly text-only MSA [Q2, Q92], aligning with the deployment's text-only modality [A4] and MSA target [elicitation]. Multimodal content was discarded [Q37] and contextual passages preserved where needed [Q38]. However, the benchmark's MSA is fluent native-speaker register from school exams, while the deployment's users are non-native learners with basic-to-intermediate MSA proficiency, likely producing grammatically imperfect or code-switched Arabic [elicitation, expected_arabic_proficiency]. This signal-distribution gap is structural and undocumented as a design trade-off in the paper. The benchmark also notes that real-world Arabic LLM exposure includes dialectal Arabic, which is excluded here [Q92]. One minor data-quality issue observed: an Arabic Language Grammar item with English question stem [DATASET-D16] inconsistent with the stated Arabic-only design [Q30].

ID	Evidence
Q2	“Our data comprises 40 tasks and 14,575 multiple-choice questions in Modern Standard Arabic (MSA) and is carefully constructed by collaborating with native speakers in the region.” (p.1)
Q92	“The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectical Arabic.” (p.9)
Q37	“During manual data scraping, workers were instructed to include only questions accompanied by an answer key, and to discard questions containing multi-modal information (e.g., images, videos, or tables).” (p.4)
Q38	“If a question had additional contextual information (e.g., a passage referenced by several questions), the context was required to be included with each corresponding question.” (p.4)
Q30	“When designing ArabicMMLU, we excluded questions in English and only included questions in Arabic.” (p.3)
D16	In the following Quranic verse, what is the correct parsing of the word ـكَ— Grammar question with English question stem — unexpected language mixing in nominally Arabic benchmark

Strengths

- Text-only MSA format directly aligns with deployment's text-based modality [Q2, Q93, A4]
- Multimodal content systematically excluded; contextual passages preserved for dependent questions [Q37, Q38]
- Prompt language variants (Arabic/English) tested, providing some signal on cross-lingual prompt sensitivity [Q47]
- Uniform MSA register confirmed across all sampled configs [DATASET-D29, DATASET-D15]

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Q47	“We initially conducted experiments with four settings: (1) Arabic prompt and Arabic alphabetic output, (2) Arabic prompt and English (i.e. Latin script) alphabetic output, (3) English prompt and Arabic alphabetic output, and (4) English prompt and English alphabetic output.” (p.5)
D29	في الآية القرآنية ﴿إِنَّ الَّذِينَ آمَنُوا وَعَمِلُوا الصَّالِحَاتِ﴾، ما هو الإعراب الصحيح لكلمة إنّ: حرف توكيد ونصب — Arabic grammar parsing using Quranic verse — MSA grammar directly relevant to deployment
D15	يعود تاريخ العرب إلى آلاف السنين... سُمِّيَ العدنانيون بعرب [فراغ]. الشمال — Pan-Arab ethnolinguistic history reading comprehension — culturally grounded, no country marker

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution mismatch: benchmark assumes fluent native-speaker MSA from school exams [Q2], while deployment users produce learner-register MSA potentially with grammatical imperfection and code-switching [elicitation expected_arabic_proficiency, languages.user_input_register]. No dedicated benchmark for non-native Arabic robustness identified [WEB-6, gap_id 5].

IF-2: Infrastructure supports text-only Arabic input; RTL rendering requirement noted [writing_systems]. No infrastructure barriers identified for tourists/expats with personal devices [infrastructure_notes].

IF-3: Domain-specific form differences: dialectal Arabic excluded from benchmark [Q92] but is common in real-world MENA usage; learner code-switching with Latin script likely [writing_systems] but not represented.

IF-4: Form mismatch: learner-register input vs. fluent-MSA evaluation creates undocumented signal-distribution gap. One data-quality artifact: English question stem in nominally Arabic-only benchmark [DATASET-D16].

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WEB-6	AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation (https://arxiv.org/abs/2510.13430)
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Information Gaps

- No empirical evaluation of model performance on learner-register or code-switched Arabic input

Requires Expert Verification

- Whether tourist/expat learner Arabic deviates from benchmark MSA in ways that materially affect downstream system behavior

Remediation

Gap 1: Benchmark MSA register does not represent learner-register, code-switched Arabic produced by non-native users [Q92, elicitation expected_arabic_proficiency]

Recommendation: Construct a small bespoke evaluation set of learner-register and code-switched Arabic queries paraphrasing ArabicMMLU items, and measure performance degradation; this addresses the documentation gap noted in [WEB-6] where no learner-register Arabic benchmark exists.

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